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Interpretable deep learning based text regression for financial prediction

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Abstract

Text regression is an important task in natural language processing (NLP), which aims to predict continuous numerical values associated with text. Previous work focused on linear text regression requiring manual feature selection for financial prediction. Recently, non-linear text regression through neural network models has become a trend. However, most models rely only on convolutional neural networks (CNN) and suffer from insufficient interpretability. In this paper, we propose a deep neural network model named EM-CBA for text regression and further interpret the model. The proposed model is powered by word EMbedding, CNN, Bidirectional long short-term memory (Bi-LSTM) and Attention mechanism. The proposed EM-CBA takes financial report texts as input and predicts a financial metric named return on assets (ROA). We conduct comprehensive experiments on a dataset about the reports of enterprises. Experimental results show that the proposed model provides more accurate predictions of enterprises' metrics than previous convolutional neural network models and other classical models. The validity of each module of the model is also verified. Finally, we demonstrate a way of performing analysis in words change and results errors to intuitively interpret the effect of different text inputs on the model. The analysis demonstrates that the model is able to use information about sentiment words to analyse their associated contexts to revise the predictions.

KEYWORDS
deep learning, financial prediction, natural language processing, text regression

1 | INTRODUCTION

Text regression involves predicting continuous numerical metrics associated with a given textual information. With the continuous development of deep learning, it has become a popular trend to study text regression problems through neural network models. Text regression techniques have been successfully effective in areas such as movie revenue Bitvai and Cohn (2015); Joshi et al. (2010), author age Nguyen et al. (2011), user voting intention Lampos et al. (2013), online reviews Ngo-Ye and Sinha (2014), and so forth.

Previous financial forecasting works were mainly based on numerical analysis methods on historical data, including risk prediction Qi et al. (2014), return prediction Baker et al. (2006); Li et al. (2013), default prediction Chen and Wu (2014); Duffie et al. (2007), stock price prediction Avramov and Chordia (2006); Grinblatt and Moskowitz (2004); Paye (2012), and so forth. Recently, techniques such as machine learning Ghosh et al. (2018); Kamruzzaman et al. (2022); Song et al. (2010), deep learning Alaminos et al. (2022); Kim and Ahn (2012); Zhou et al. (2022) and graph neural networks Wu et al. (2022); Xia et al. (2022) have been applied in financial forecasting. Only focusing on historical numerical data may have certain limitations. Therefore, text information as an important information should be widely adopted in financial forecasting. In financial texts,

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financial reports are normative and authoritative texts that reflect the overall operating conditions of the company. Scholars predict stock return volatility through financial reports based on text regression Dereli and Saraclar (2019); Kogan et al. (2009); Yeh et al. (2020). Linear text regression methods require artificial feature selection. The selection of features is subjective, or features may be based on misinformation. Although non-linear text regression models outperform linear text regression, the models lack sufficient interpretability. More extensive experiments are needed to demonstrate the feasibility of non-linear text regression models.

This paper investigates the text regression problem by predicting a corporate financial indicator from financial report texts. In particular, we develop a deep neural network model named EM-CBA which automatically learns the information of the text through training. The proposed model provides a new perspective on linking financial report text information and return on assets (ROA). As one of the comprehensive metrics to measure the profitability of enterprises Guoci and Liansheng (2005), ROA is predicted by the proposed model inputting financial report text information. In addition, we combine the specific report text to further analyse the interpretability of the model.

The overall of the proposed network includes embedding layer, local information extraction layer, global information extraction layer and output layer. The input text will be converted into word embedding in the embedding layer. Convolutional neural network (CNN) with different kernel sizes extract word embedding to form local text information. After that, max-over-time pooling operation is applied to different local text information. Different local text information will be concatenated together as the output of the local information extraction layer. The global information extraction layer consists of a Bidirectional Long Short-Term Memory (Bi-LSTM), a Multi-head self-attention mechanism Vaswani et al. (2017) and a Add & Norm Vaswani et al. (2017) layer. The global information extraction layer extracts global text information from local information extraction layer and evaluates the contribution of each part of information. At last in output layer, global text information will be sent into two fully connected layers to output a result of prediction.

The contributions of this paper can be summarized from the following aspects:

1. We propose a deep neural network which only relies on text content, which are annual reports and semi-annual reports of 24 listed marine enterprises in Guangdong Province from 2000 to 2021. Compared with CNN-based text regression models and classical models, the proposed model EM-CBA performs better in ROA prediction.
2. The proposed model is more lightweight than the encoder of transformer Vaswani et al. (2017) with attention mechanism. EM-CBA outperforms transformer in terms of MAE and Spearman's coefficent while achieving a shorter training time.
3. We interpret the effect of different text inputs on EM-CBA model by performing analysis in words change and results errors. The analysis shows that our proposed model can effectively utilize text information.

The remainder of this paper is given as follows. Related work is discussed in Section 2. In Section 3, we define the problem statement. Section 4 describes the proposed model in detail. We present the experimental results and model analysis in Section 5 and Section 6. Finally, our research is concluded in Section 7.

2 | RELATED WORK

This paper applies the method of text regression to predict the ROA of enterprises. This section will review the researches related to ROA and introduce the method of text regression.

2.1 | ROA prediction

As one of the most important and popular financial metrics, ROA was widely adopted in the investigation of financial metrics. For example, Siminica et al. (2012) conducted a statistical investigation of the relationship between ROA and 24 financial indicators. The study was designed to analyse the correlation of ROA with other indicators on a dataset of 40 Romanian companies listed on the Bucharest Stock Exchange. Suardana et al. (2018) analysed the impact of six corporate indicators on ROA and changes in profits for Rural Banks in Bali Province. In studies of enterprise analysis, ROA had been utilized to predict firm failure. Beaver (1966) treated ROA as one of the six financial ratios that predict firm failure. Similarly, Altman (1968) also considered ROA as one of five financial ratios that predict corporate bankruptcy. ROA was also considered as a criterion for evaluating a company's financial performance. Rostami et al. (2016) analysed the impact of corporate governance components on corporate financial performance through data from about 469 companies over a period of 7 years. Specifically, they analysed the impact of ROA and Return On Equity (ROE) through six internal components of the corporate governance system. In addition, ROA is widely employed to reflect the profitability of a company. Heikal et al. (2014) treated ROA as one of the dependent variables to analyse the profit growth of automotive firms listed on the Indonesian stock exchange. Ahmed et al. (2020) conducted an empirical study on the issue of corporate governance predicting corporate profitability. They measured a company's profitability by ROA and ROE. Analogously, Choi et al. (2011) explored the issue of dividend

changes predicting firm profitability. ROA and ROE were employed to reflect the profitability of firms in the paper. Mundt et al. (2020) exploited ergodicity to predict profitability which is measured by the annual ROA.

In summary, ROA is an effective financial indicator for financial analysis. Scholars have studied ROA by various numerical analysis methods. However, there seems to be little research to study ROA through textual information. This paper links corporate financial report text information with ROA. We present a novel approach to applying textual information to financial forecasting problems.

2.2 | Text regression

There are a large number of subtasks in the NLP field Zhang and Chen (2022); Lai et al. (2022). Text regression is a subtask in the field of NLP; it can be divided into linear regression and non-linear regression.

Most early studies focused on linear text regression. For example, Kogan et al. (2009) defined a text regression task about predicting financial risk from annual report texts. They built regression models to predict stock return volatility from the text of a company's 10-K report. Based on a linear text regression method, Joshi et al. (2010) used the textual features of movie reviews and meta-features of movies to predict the total revenue and average revenue per screen of a movie's opening weekend. Meanwhile, the effectiveness of linear text regression has been validated in predicting author age Nguyen et al. (2011), predicting user voting intent Lampos et al. (2013) and tracking influenza epidemics Lampos and Cristianini (2010). Although the method of linear text regression had been validated in various fields, the selection of indicators and the determination of weights are subjective, or they may be based on inaccurate information.

With the development of deep learning, non-linear text regression had also been applied in the field of financial risk Yeh et al. (2020) and online reviews Ngo-Ye and Sinha (2014). The non-linear text regression methods do not require artificial selection of features, but build a model to train a dataset. Recently, non-linear text regression combined with neural networks, especially CNN, had been employed in research. On the basis of previous studies, Bitvai et al. Bitvai and Cohn (2015) presented a non-linear method based on a deep convolutional neural network to predict the future box-office takings of movies and showed that non-linear method works better than the linear method. Dereli and Saraclar (2019) proposed a convolutional neural network model and applied transfer learning to forecast stock return volatility from annual reports. Paetzold and Specia (2017) proposed and tested the text regression model using deep convolutional neural networks and multi-layer perceptron on two text regression tasks. Furthermore, Ye et al. (2021) developed a text regression-based model with hierarchical structure and multi-head attention mechanism for specific news domain push. Research on text regression models with RNN or attention mechanisms is absent. Although these models work well, they do not have strong interpretability. It is necessary to demonstrate the feasibility of the model through more extensive experiments.

In this paper, deep learning techniques in NLP are adopted for non-linear text regression. Concretely, EM-CBA is proposed to predict enterprises' ROA from financial report texts. We not only verify the performance of the model through experiments, but also analyse the effect of the model on different texts.

3 | PROBLEM STATEMENT

In this section, ROA as the enterprise metric is defined. Following that, we will introduce the dataset which is used in this paper. Finally, we will describe the evaluation measures of the proposed model.

3.1 | Return on assets

In this paper, ROA is considered as a predicted enterprise metric. ROA is not only a very useful ratio when analysing an enterprise's profitability, but also a reliable indicator of enterprise's competitiveness. It can be computed by Needles et al. (2013)

$$\mathrm{ROA} = \frac{\text{Net income}}{\text{Average total assets}} \times 100\%. \tag{1}$$

ROA is an index of how much net income is generated per unit of assets. It is a comprehensive index, and its level directly reflects the enterprise's competitive strength and development ability. It is worth noting that ROA varies greatly across different industries, which is suitable for comparison among different competitive enterprises in the same industry. Therefore, in this paper, we take ROA as an indicator to measure different enterprises in the marine industry of Guangdong Province.

3.2 | Dataset

In this paper, the dataset contains text and ROA data of 24 listed marine enterprises in Guangdong Province from 2000 to 2021. We first collect the annual and semi-annual reports of each enterprise from Tide Information website[i] and Chinese Finance website.[ii] Secondly, the section of Board Report or Management's Discussion and Analysis is extracted from reports to use instead of the complete reports. Finally, we collect relevant data for each enterprise from the Financial Choice Terminal to calculate the ROA.

3.3 | Evaluation

Mean absolute error (MAE) is selected as the main evaluation metric

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |\, y_i - \widehat{y}_i \,|, \tag{2}$$

where y_i is the real ROA value, $\widehat{y}_i$ is the predicted ROA value and N is the number of y_i. MAE is chosen as the loss function of all training models.

Spearman's rank correlation coefficient is a metric used to evaluate the ranking performance of a model. It can be calculated by the Pearson's correlation coefficient of the ranking of two sets. The ranking of a set can be generated by ascending and enumerating the values in the set. Assuming $\mathrm{rank}(X)$ and $\mathrm{rank}(Y)$ are the ranking of set X and set Y respectively, the formula of Spearman's rank correlation coefficient can be

$$\rho_{\mathrm{rank}(X),\ \mathrm{rank}(Y)} = \frac{\mathrm{cov}(\mathrm{rank}(X),\ \mathrm{rank}(Y))}{\sigma_{\mathrm{rank}\ (X)}\sigma_{\mathrm{rank}\ (Y)}}, \tag{3}$$

where $\mathrm{cov}(\mathrm{rank}(X),\ \mathrm{rank}(Y))$ represents the covariance of the rank variables. $\sigma_{\mathrm{rank}(X)}$ and $\sigma_{\mathrm{rank}(Y)}$ represent the standard deviations of the rank variables. In this work, we use real ROA values and predicted ROA values to calculate Spearman's rank correlation coefficient which is used to evaluate the ranking performance of different models.

4 | MODEL

In this section, we introduce the proposed EM-CBA model. The overall architecture of the model is illustrated in Figure 1 which consists of embedding layer, local information extraction layer, global information extraction layer and output layer. Through the embedding layer, the local information extraction layer encodes words information to form the local information of the text. Next, the global information of text will be formed in the global information extraction layer. At last, we feed it into the output layer to output the prediction results. We will present the design of each layer at length in the followings.

4.1 | Embedding layer

Word embedding is a method of representing words in vectors to embed syntactic and semantic information, which maps words into low-dimensional vectors. In this work, we adopt pretrained word embeddings Li et al. (2018). They provided many types of Chinese word embedding, but we use one of them. The word embedding we use was trained by a corpus of financial news collected from several websites with a vector dimension of 300. The embedding of all words forms a word embedding matrix E with size $|U| \times d$, where $|U|$ is the number of all words and d is the word embedding dimensionality. The matrix E is a parameter to be learned.

The input tokens will be looked up in the word embedding matrix E in order to obtain a distributed representation of the input text. Through the embedding layer, input text can be transformed into a document matrix W with size $m \times d$, where m is the max length of input text. If the length of input text is less than m words, the document matrix will be padded with zero vector. Besides, the unknown word vectors in the document matrix will be randomly initialized.

4.2 | Local information extraction layer

In this layer, CNN with multiple variant window sizes is used to extract local information of the document matrix W generated by embedding layer. Let $W = [w_1, w_2, \ldots, w_m]^T \in \mathbb{R}^{m \times d}$, where $w_i \in \mathbb{R}^{1 \times d}$ is the embedding of the i-th word in the input text. We apply convolution operations with

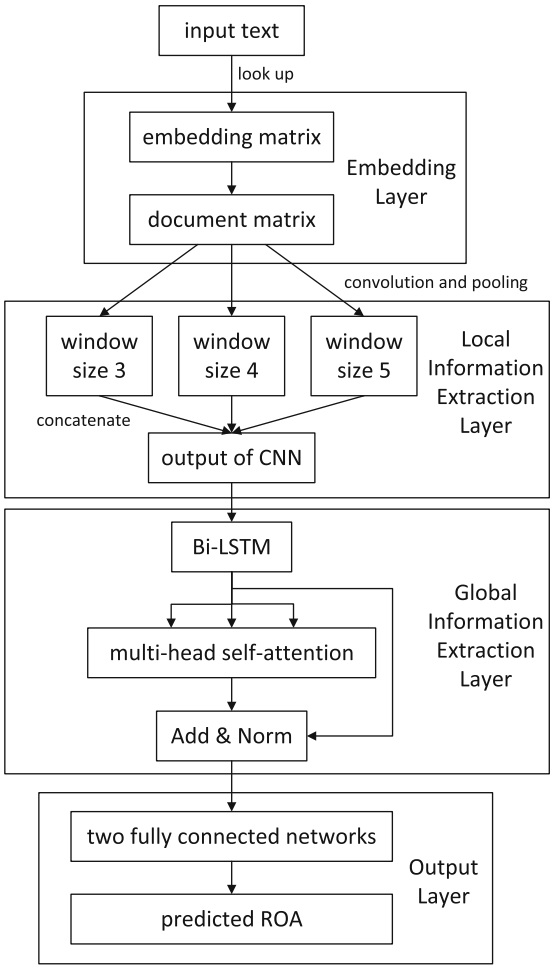

FIGURE 1 The overall architecture of EM-CBA.

filters $f^s \in \mathbb{R}^{s \times d}$ containing window sizes 3, 4, and 5 to the document embedding matrix $w_{1:m}$, where s is the window size of the filter. Let the number of filters for each window size be n. A convolution feature p_{ij} is generated by

$$p_{ij} = \mathrm{ReLU}\left(f_j^s * w_{i:i+s-1} + b\right), \tag{4}$$

where ReLU Xu et al. (2015) is rectified linear unit which is employed as non-linear activation function, $f_j^s (1 \le j \le n)$ is the j-th filter, $*$ is the convolution operator and $b \in \mathbb{R}$ is bias. Since the number of filters for each window size is n, each sub-matrix $w_{i:i+s-1}$ generates n convolution features. Filters are applied for each position of document embedding to obtain the whole convolution feature map $P = [p_1, p_2, \ldots p_{m-s+1}]^T \in \mathbb{R}^{(m-s+1) \times n}$, where $p_i \in \mathbb{R}^{1 \times n} (1 \le i \le m - s + 1)$.

In the next step, we apply max-over-time pooling operation on the convolution feature map P to retain the most important features. Max-over-time pooling operation is applied sequentially to k rows of the convolution feature map instead of the complete map. The maximum value of the column direction will be retained. After the above operation, there are r rows of output $\widetilde{C} = [c_1, c_2, \ldots c_r]^T \in \mathbb{R}^{r \times n}$,

$$c_i = \max\left(p_{ik:i(k+1)-1}\right), \tag{5}$$

where $c_i \in \mathbb{R}^{1 \times n}$. Simultaneously, $\widetilde{C}$ can be represented by n columns, $\widetilde{C} = [g_1, g_2, \ldots g_n] \in \mathbb{R}^{r \times n}$, where $g_i \in \mathbb{R}^{r \times 1}$. Figure 2 demonstrates the convolution and pooling process for a filter with a window size of 3. Three types of filters with different window sizes can produce different features and they are concatenated as the output $C \in \mathbb{R}^{3n \times r}$ of the local information extraction layer.

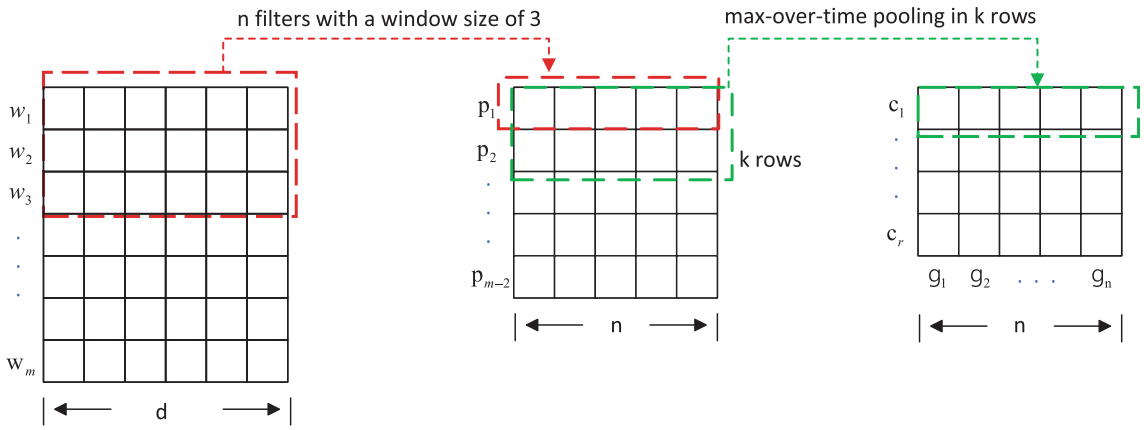

FIGURE 2 Convolution and pooling process for a filter with a window size of 3.

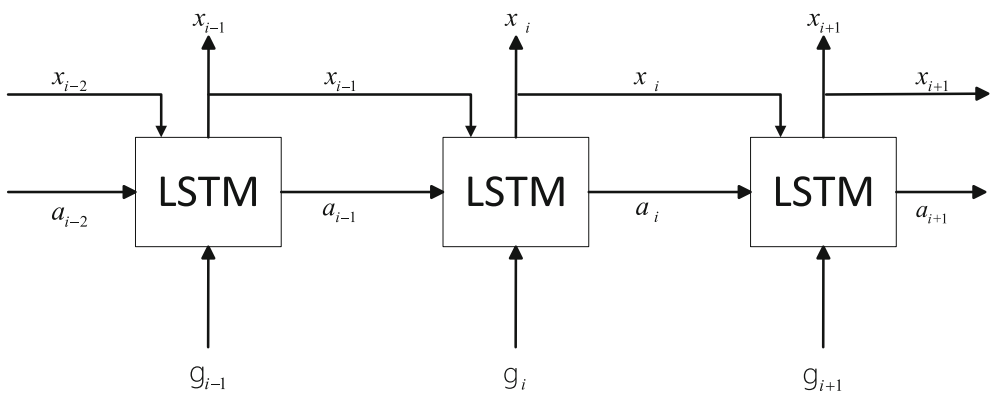

FIGURE 3 A schematic diagram of unidirectional long short-term memory (LSTM) units.

4.3 | Global information extraction layer

To process the output from the local information extraction layer, we adopt a Bi-LSTM, a multi-head self-attention mechanism and a Add & Norm layer to further extract global information. LSTM Hochreiter and Schmidhuber (1997) is a modification of RNN, which has special adaptive gating mechanism that can avoid gradient vanishing or exploding problem to a certain degree. Specifically, LSTM has three gating mechanisms: forget gate Γ_f, update gate Γ_u, output gate Γ_o. Figure 3 is a schematic diagram of unidirectional LSTM units. The state x_{i-1} and the state a_{i-1} are generated from previous LSTM unit. Through them and the current input g_i, we can complete the calculation of an LSTM unit:

$$\Gamma_f = \sigma(W_{fg}g_i + W_{fx}x_{i-1} + b_f), \tag{6}$$

$$\Gamma_u = \sigma(W_{ug}g_i + W_{ux}x_{i-1} + b_u), \tag{7}$$

$$\Gamma_o = \sigma(W_{og}g_i + W_{ox}x_{i-1} + b_o), \tag{8}$$

$$\widetilde{a}_i = tanh(W_{ag}g_i + W_{ax}x_{i-1} + b_a), \tag{9}$$

$$a_i = \Gamma_u\widetilde{a}_i + \Gamma_f a_{i-1}, \tag{10}$$

$$x_i = \Gamma_o a_i, \tag{11}$$

where W_{fg}, W_{fx}, W_{ug}, W_{ux}, W_{og}, W_{ox}, W_{ag}, W_{ax}, b_f, b_u, b_o, and b_a are weight matrix and parameters to be learned through training. $\sigma(\cdot)$ is the sigmoid activation function that outputs values between 0 and 1, thus gates work according to values between 0 and 1. The gate fails when the value is 0, and exerts its full effect when the value is 1. Furthermore, Bi-LSTM extends opposite direction on the basis of unidirectional LSTM,

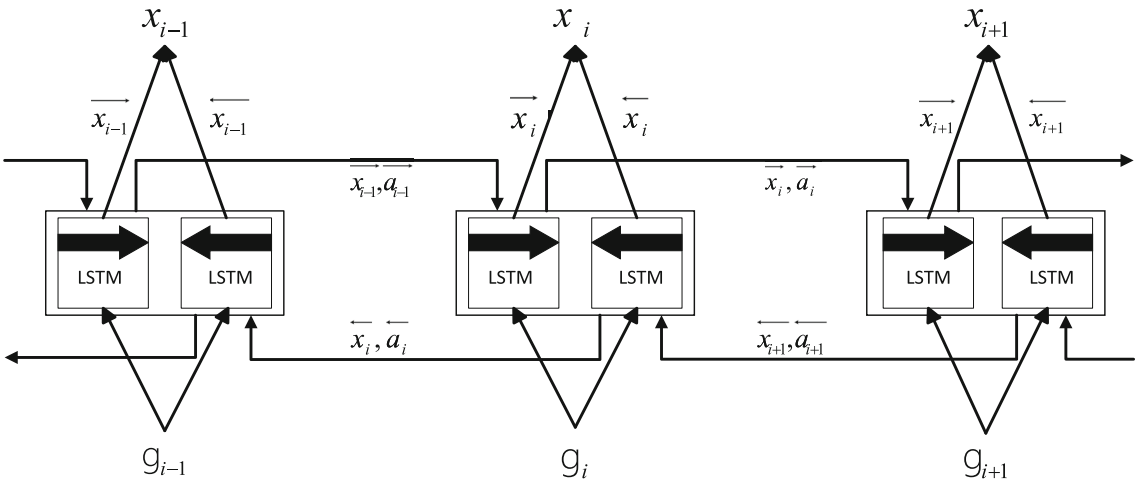

FIGURE 4 A schematic diagram of bidirectional long short-term memory (Bi-LSTM) units.

which takes full advantage of past and future information to process sequences. Figure 4 is a schematic diagram of Bi-LSTM units. The output of the Bi-LSTM consists of forward $\overrightarrow{x_i}$ and backward $\overleftarrow{x_i}$. The former is from left to right and the latter is reversed. We concatenate the two directions to get the final output of the Bi-LSTM unit:

$$x_i = \left[\overrightarrow{x_i} \oplus \overleftarrow{x_i}\right], \tag{12}$$

where $\oplus$ means element-wise sum. The output formed by all Bi-LSTM units is $X = [x_1, x_2, \ldots, x_{3n}]^T$.

Multi-head self-attention mechanism was first proposed for translation tasks and achieved good results. The method for calculating self-attention scores refers to the idea of querying values in a table by keys, where the queries, keys, and values are all vectors. We calculate the dot product by the query vector and all key vectors. Following that, all dot products are divided by $\sqrt{d_K}$ (d_K is the dimension of key vector) and softmax function is utilized to calculate the weights on the value vectors. In practice, we compute a set of queries, keys and values simultaneously and pack them each into a matrix. We obtain the query matrix $Q \in \mathbb{R}^{3n \times r}$, key matrix $K \in \mathbb{R}^{3n \times r}$ and value matrix $V \in \mathbb{R}^{3n \times r}$ by multiplying each of three parameter matrices W_Q, W_K, and W_V with the output of Bi-LSTM X. The formula for using the matrix to compute can be expressed as

$$\text{Attention}\ (Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right) V, \tag{13}$$

where K^T is the transpose of K. The number of heads h is a hyperparameter. We divide the three matrices Q, K, V equally into h parts by rows and execute Equation (13) for each part. Then all parts are concatenated into a matrix H. The output $\widetilde{Z}$ of multi-head self-attention mechanism is generated by multiplying H with a parameter matrix W_O. For regularization, $\widetilde{Z}$ will be added to the output X of Bi-LSTM and layer normalized in the Add & Norm layer. Eventually, $Z \in \mathbb{R}^{3n \times r}$ is fed into the final layer of the model as the output of the global information extraction layer.

4.4 | Output layer

There are two fully connected networks in this layer. In these two fully connected networks, the output Z of the upper layer is sequentially multiplied with two parameter matrices W_1, W_2 to change the dimensionality. The result of the multiplication is the final prediction $y \in \mathbb{R}$ of the model. The loss function calculates the error between the predicted value and the true value. The model is trained and optimized continuously according to the value of loss function through back propagation.

5 | EXPERIMENT

This section first describes experimental setup including data preprocessing, parameter settings and baselines. Later, we present and analyse the results of comparative experiments of the model.

5.1 | Experimental setup

5.1.1 | Preprocessing

Before the input text is fed into the model, the text will be tokenized by splitting text into words through the Chinese word splitting tool jieba, removing punctuation and stop words and replacing # with numerical values. After completing the above operations, we shuffle the dataset into training, validation, and test sets in a ratio of 7:1.5:1.5. The characteristics of the dataset are shown in Table 1.

5.1.2 | Baselines

To estimate the validity, we compare the proposed EM-CBA with the following baselines, in which the first two baselines are CNN previously used for text regression and the rest are classical models. The baselines are described as follows:

1. CNN-F Dereli and Saraclar (2019): A CNN for financial text regression that takes 10-K company reports as input and outputs a prediction of stock return volatility. Specifically, the CNN has three types of window sizes 3, 4, and 5 and max-over-time pooling layer. After the convolution and pooling operations, two fully connected networks output the predictions of the model.
2. CNN-T Bitvai and Cohn (2015): A CNN for text regression that takes movie critics and movie attributes as input and outputs a prediction of future box office revenue. Specifically, the CNN has three types of window sizes 1, 2, 3 and max pooling layer. After the convolution and pooling operations, three fully connected networks output the predictions of the model.
3. Bi-LSTM: A model employs Bi-LSTM that has 50 hidden states in one direction. This is followed by the predictions output by two fully connected networks.
4. Bi-GRU: A model employs Bi-GRU that has 50 hidden states in one direction. This is followed by the predictions output by two fully connected networks.

5.1.3 | Parameter settings

The models in this paper are implemented by using the deep learning pytorch framework. The models are trained by mini-batch approach and are optimized by the Adam optimizer Kingma and Ba (2015). Besides, dropout technique Hinton et al. (2012) is applied to alleviate overfitting. All weight matrices are randomly initialized and trainable in this experiment. We have used different parameters to evaluate the baseline approaches. Specifically, we adjusted the length of the fixed text within $\{2000, 3000, 4000, 5000\}$, the size of the learning rate within $\{0.01, 0.001, 0.0001\}$, the size of the batch within $\{4, 8, 16, 32\}$ and the probability of dropout within $\{0.3, 0.5, 0.7\}$. After extensive experiments, we tested various combinations of hyperparameters and finally determined the optimal one. We used a uniform combination of hyperparameters to fairly test the performance of the baseline models. Table 2 shows the specific experimental parameter settings. The number of filters for CNN is set to 100. The number of heads in multi-head self-attention mechanism is set to 5.

5.2 | Results

The results of comparison between different models and the comparison among various alternative configurations of EM-CBA are presented in this subsection.

TABLE 1 Characteristics of the dataset.

	Train	Validate	Test	Total
# Texts	415	90	89	594
# Average text length	9416	10,155	8758	9429
# Words per text	4017	4307	3770	4024

TABLE 2 Experimental parameter settings.

Parameter	Fixed text length	Learning rate	Batch-size	Dropout	Epoch
Value	3000	0.001	8	0.3	50

5.2.1 | Comparison between different models

Table 3 presents the results of comparison between different models on the test set. MAE is used to reflect the error between the model prediction result and the true value. Spearman's rank correlation coefficient reflects not only how closely the predicted values are related to the true values, but also the ranking performance of each model. The higher Spearman's rank correlation coefficient, the closer the trend of the predicted value is to the true value, and the larger the proportion of data changes captured by the model. We can see that the EM-CBA achieves the best performance, both in terms of MAE and Spearman's rank correlation coefficient. Compared with the two CNN models and the two classic models, our model drops by 29.77%, 10.15%, 11.32% and 23.70% in MAE, respectively. Meanwhile, Spearman's rank correlation coefficient is also improved by 37.46%, 11.65%, 11.99% and 17.00% respectively. It can be seen that the models that output the prediction results through a single neural network followed by fully connected networks cannot achieve the best results. They may not adequately extract the feature information of the text. The layered neural network structure allows for more adequate extraction of information from the text.

We tested the training time of the models on an RTX3090 graphics card. CNN-F and CNN-T use only CNN and pooling operation to output prediction results. Compared to RNN, the parallelization calculation of CNN is better. Moreover, CNN and pooling operation can greatly reduce the dimension of the matrix, thus reducing the training time. However, compared with the hierarchical structure of EM-CBA, the prediction performance of CNN-F and CNN-T is not as good as EM-CBA. Bi-LSTM and Bi-GRU do not greatly reduce the dimensionality of the matrix and do not have pooling operations, so they have a longer training time.

5.2.2 | Comparison among various alternative configurations

To verify the effectiveness of the network structure, we compare various configurations of EM-CBA from four aspects: the setting of word embeddings, the amounts of window size in CNN, the selection of RNN variants, and the setting of attention mechanism. The results of comparison among various alternative configurations of EM-CBA on the test set is shown in Table 4.

TABLE 3 Comparison between different models.

Model	MAE	Spearman's rank correlation coefficient	Training time (s)
CNN-F	2.0982	0.5558	942.66
CNN-T	1.6400	0.6843	**546.61**
Bi-LSTM	1.6617	0.6822	4125.52
Bi-GRU	1.9312	0.6530	3557.37
EM-CBA	**1.4736**	**0.7640**	2225.79

Note: Boldface shows the best result among presented models for the corresponding column.

TABLE 4 Comparison among various alternative configurations.

Model description	MAE	Spearman's rank correlation coefficient
Randomly initialize word embeddings	1.8948	0.6625
Use pretrained word embedding	**1.4736**	**0.7640**
Filters with window size 3	1.7865	0.6618
Filters with window size 3,4	1.4920	0.7324
Filters with window size 3,4,5	**1.4736**	**0.7640**
Filters with window size 3,4,5,6	1.5315	0.7264
Without CNN	1.6996	0.6618
Bi-LSTM	**1.4736**	**0.7640**
Bi-GRU	1.8026	0.6924
Without RNN variants	2.0364	0.6248
With attention	**1.4736**	0.7640
Without attention	1.6659	**0.7650**

Note: Boldface shows the best result among corresponding various configurations.

We can observe that using pretrained word embeddings compared to randomly initialized word embeddings improves the MAE and Spearman's rank correlation coefficient by 22.23% and 15.32%, respectively. This demonstrates the effectiveness of using pretrained word embeddings in EM-CBA.

The ideal window size for the filter is 3, 4, and 5. Filters with a single category window size are slightly less effective, but filters with too many category window sizes are equally ineffective. It has been tested that taking a smaller window size $s < 3$ or a larger window size $s > 5$ does not help. We also test the model without CNN, but the result is not as bad as expected. It can be seen that the importance of setting the window size correctly. Appropriate window size helps filters to extract local information of the text, or it prevents the model from outputting the correct predictions.

Bi-LSTM and Bi-GRU are both variants of RNN. Bi-LSTM is able to achieve better result. The model without RNN variants is also tested as a comparison. The results show that Bi-LSTM performs well to further process the output from the local information extraction layer.

The multi-head self-attention mechanism brings 11.54% gain to MAE. We will give another example of attention mechanism in the next subsection.

5.2.3 | Comparison with extended model

The multi-head self-attention mechanism does improve the performance of EM-CBA. We further explore whether only using the attention mechanism can get better prediction results, so we train data by using the encoder of transformer which first employed multi-head attention and performed well in translation tasks. The encoder consists of positional encoding, multi-head attention, Add & Norm layer and feed-forward layers. The results compared with transformer are shown in Table 5. EM-CBA performs much better than transformer in MAE and Spearman rank correlation coefficient. Additionally, transformer takes more than five times as long to train as EM-CBA when exploiting an RTX 3090 graphics card. EM-CBA has a short training time may mainly thanks to the local information extraction layer, which uses convolutional neural networks and a max-over-time pooling operation. They can greatly reduce the dimension of the matrix, thus reducing the training time. The prediction problem in this paper is not suitable to be solved by the encoder of the transformer. However, we can achieve better results by introducing its multi-head attention mechanism.

6 | ANALYSIS

In this section, words change and prediction results error of the report texts will be analysed to intuitively interpret the effect of different text inputs on EM-CBA.

6.1 | Words change analysis

We try to analyse the influence of words on the prediction results of EM-CBA. The embedding weights of trained EM-CBA are compared with word embedding matrix before training to identify words with the largest change. The top 10 words with the largest change measured by cosine distance and Euclidean distance are presented in Table 6. "吸引 (absorb)" and "赞成 (approve)" are positive words, while "fall into (陷入)" is a negative word in Chinese financial sentiment dictionary Jiang et al. (2019). We consider "攀升(rising)" should also belong to positive words, because most words related to rising are positive words in the dictionary. There are four financial sentiment words in the two groups of top 10 most varied words. In addition to this, there are words that indicate time, place, and specific areas. We can conclude that EM-CBA can capture the sentiment information of the text to a certain extent, but it does not rely solely on sentiment information to make predictions.

In the next step, we analyse whether the impact of sentiment words on the prediction results is as expected. We sequentially set the word embeddings of four sentiment words to zero and measure the average prediction difference for each report on test set before and after the setting change. We report the impact of four sentiment words on the prediction results of the test set in Figure 5. It can be seen that three words

TABLE 5 Comparison with extended model.

Model	MAE	Spearman's rank correlation coefficient	Training time(s)
Transformer	2.7780	0.1328	11679.36
EM-CBA	**1.4736**	**0.7640**	**2225.79**

Note: Bold indicates the best result among presented models for the corresponding column.

TABLE 6 Top 10 most varied words measured by cosine distance and Euclidean distance.

Word	Cosine distance	Word	Euclidean distance
攀升 (rising)	0.3594	沥青(asphalt)	4.7530
安江 (An Giang)	0.2364	攀升 (rising)	4.6557
沥青 (asphalt)	0.2195	一二年 (1 or 2 years)	4.1533
解释 (explain)	0.2064	仓储业 (warehousing industry)	3.6275
深能源 (deep energy)	0.2055	运量 (freight volume)	3.5948
吸引 (absorb)	0.2038	赞成 (agree)	3.5848
看 (look)	0.1988	期初 (beginning of the period)	3.4231
日邮 (Daily Mail)	0.1920	不达 (not up to)	3.4157
陷入 (fall into)	0.1789	科啸 (Ke Xiao)	3.3508
东南亚 (Southeast Asia)	0.1753	东南亚 (Southeast Asia)	3.3479

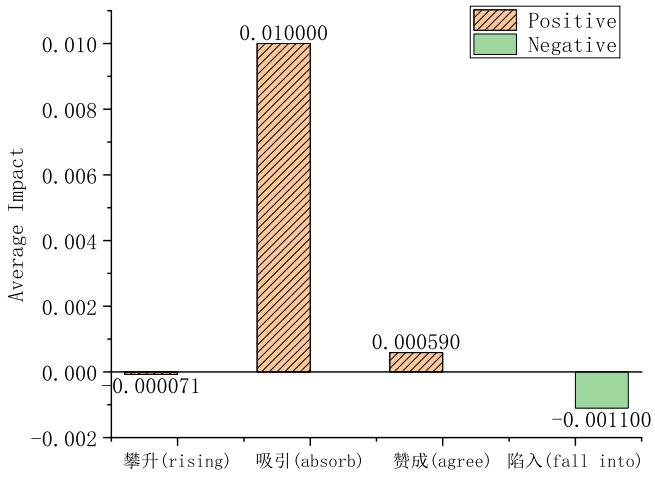

FIGURE 5 Four sentiment words on the predictions in the test set.

"吸引 (absorb)", "陷入 (fall into)", and "赞成 (approve)" are as expected, where positive words can increase the value of ROA, while negative words do the opposite. However, the word "攀升 (rising)" seems to be different. We find those reports where the word "攀升 (rising)" results in the reduction on output. In the report with the most reduction in output, the descriptions are as follows:

1. "业务量的减幅主要原因是各类原材料价格, 人工成本及贷款利率等同比大幅攀升 (The decrease in business volume was mainly due to the sharp year-on-year rise in the prices of various raw materials, labour costs and loan interest rates.)".
   a. Other reports have similar descriptions:
2. "加上航油, 航材价格的不断攀升制约了业绩的提升发展 (In addition, the rising prices of aviation fuel and aviation materials have restricted the improvement and development of performance.)".
3. "近年来中国沿海出现劳工短缺, 劳动力成本攀升等趋势 (In recent years, there have been labour shortages and rising labour costs in China's coastal areas.)".

In these reports, the word "攀升 (rising)" is no longer a positive word due to the contextual difference. So it makes sense that the predicted ROA is reduced. It shows that the proposed EM-CBA can correctly use sentiment words and comprehensively consider the context of words.

6.2 | Results errors analysis

Errors between predicted ROA and true ROA in test set are analysed in this part. The scatter plot of predicted ROA and true ROA is presented in Figure 6. The distance between two points in the vertical axis direction indicates the error value. It can be observed that within the ROA value

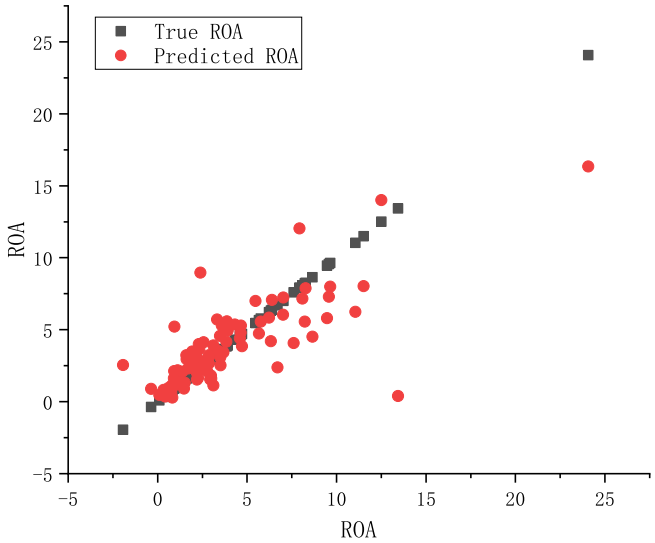

FIGURE 6 Scatter plot of predicted return on asset (ROA) and true ROA in the test set.

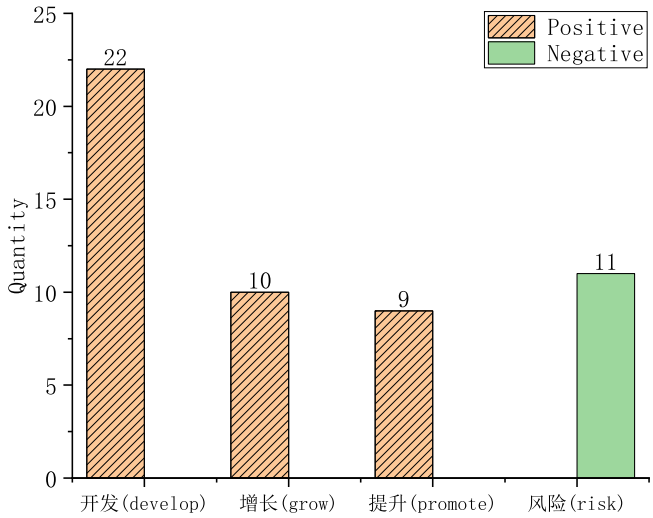

FIGURE 7 The quantity of sentiment words (the error between predicted return on asset (ROA) and true ROA is +4.5044).

ranging from 0 to 5, EM-CBA predicts the best. However, EM-CBA predicts less well in the range of ROA less than 0 and greater than 5. For example, the true ROA value is less than 0 but the predicted ROA value is greater than 0 or the error between them is large.

In addition, it is shown in Figure 6 that the maximum true ROA is 24.0706, while the predicted ROA is 16.3495. The gap between them is 7.7211. The minimum true ROA is −1.9390, while the predicted ROA is 2.5654. The gap between them is 4.5044, and the numerical signs between them are opposite. We infer that their large errors may be due to the small amount of data. Because the number of ROA values less than 0 and greater than 20 only accounts for 3.37% and 0.84% of dataset. The maximum error between the two points is 13.0187, where the real ROA is 13.4319 and the predicted ROA is 0.4132. Due to the large error in the prediction results of these points, we continue to analyse the details of their report texts in depth.

Furthermore, we try to analyse the emotional polarity of their report texts. Sentiment words from the list of the top 50 most occurring words are extracted in Figure 7, Figure 8 and Figure 9. In Figure 7, we find that the report text with +4.5044 error between predicted ROA and true ROA has more positive words, which may affect the prediction of the EM-CBA. As can be seen in Figure 8, the number of positive and negative words contained in the report text with predicted and true ROA errors of −7.7211 is similar. Therefore, the information of sentiment words may not help EM-CBA to make the correct prediction. For the report text with an error of −13.0187 between the predicted ROA and the true ROA, there are only a large number of negative words in its top 50 most frequent word list. The prediction of the EM-CBA may be seriously disturbed by too many negative words.

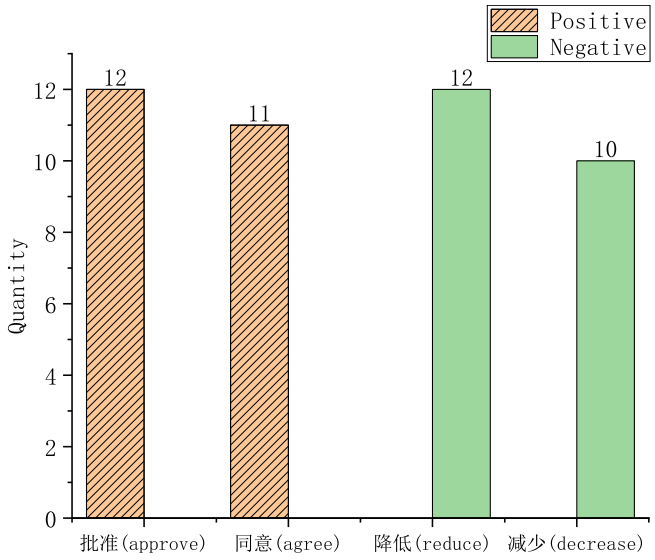

FIGURE 8 The quantity of sentiment words (the error between predicted return on asset (ROA) and true ROA is −7.7211).

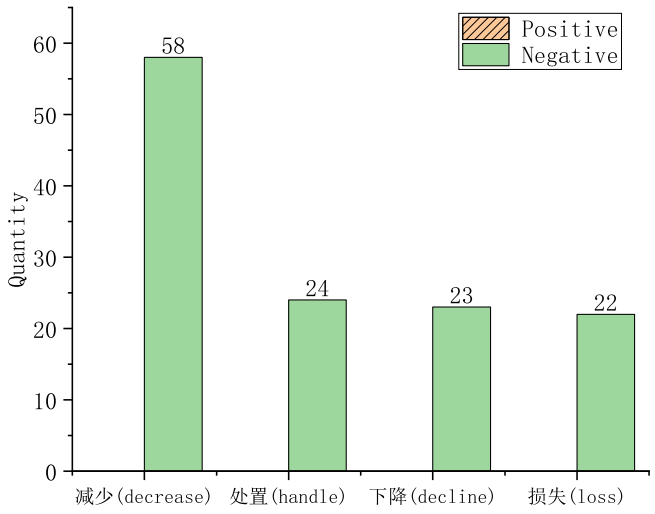

FIGURE 9 The quantity of sentiment words (the error between predicted return on asset (ROA) and true ROA is −13.0187).

Figure 10 illustrates absolute value of maximum result error for 24 enterprises in the test set. The absolute value of maximum result error for most enterprises is between 0 and 5. However, there are 3 enterprises with larger values. The maximum error value for company with stock code 000088 is 7.7211, for company with stock code 300424 is 6.5776, and for company with stock code 600685 is 13.0187. The emotional polarity of report texts with error values of 7.7211 and 13.0187 has been analysed in Figures 8 and 9 above. We proceeded to analyse the emotional polarity of the report text with an error value of 6.5776. In Figure 11, we find only positive words in the list of the top 50 most frequently occurring words in the text. The prediction of EM-CBA may be influenced by positive emotional words, resulting in a higher predicted value than the true value. Although the EM-CBA can effectively utilize the information of sentiment words, sometimes sentiment words may interfere with the prediction of the model to some extent.

7 | CONCLUSION

In this paper, we investigated a text regression method for predicting the ROA metric of enterprises from their financial report texts. A deep neural network model named EM-CBA was proposed and verified on a dataset about the reports of enterprises. The experimental results demonstrate the superiority of our proposed model compared with other models available for text regression, as well as the effectiveness of the model structure. Moreover, we analysed the ability of EM-CBA from the perspective of word change and result error. The analyses show that EM-CBA

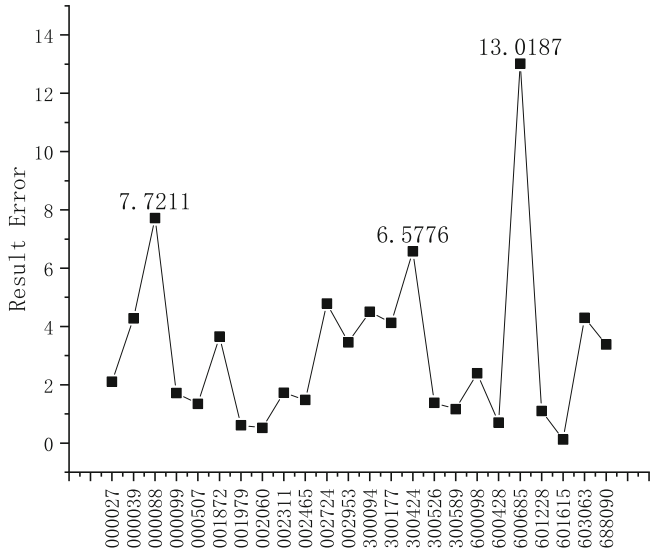

FIGURE 10 Absolute value of maximum result error for 24 enterprises in the test set.

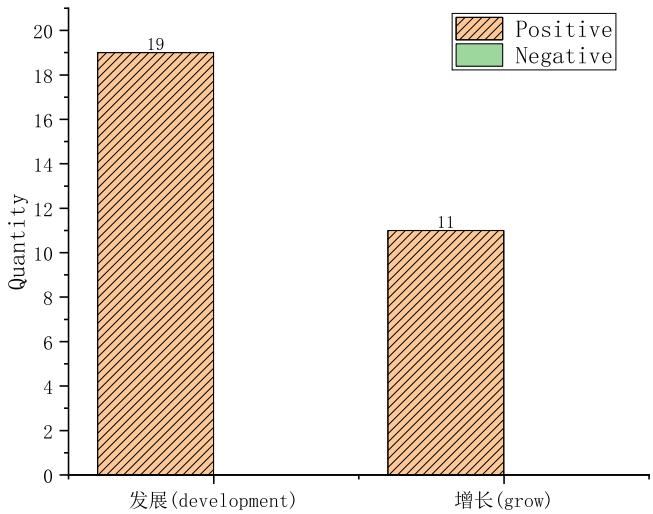

FIGURE 11 The quantity of sentiment words (the error between predicted return on asset (ROA) and true ROA is +6.5776).

cannot only successfully track the information of sentiment words, but also comprehensively consider its context. However, sometimes sentiment words may interfere with the prediction of the model to some degree. In conclusion, our approach further explores the problem of text regression through neural network models and provides new ideas for employing textual information for financial forecasting.

This work is only focused on the text portion of enterprises' reports. In the future, we will consider combining with other methods and adding enterprise historical data or other metadata to EM-CBA, which may further improve the performance of the model.

AUTHOR CONTRIBUTIONS

The study conception and design were proposed by Weiwen Zhang and Rufeng Liang. Data collection and analysis were performed by Rufeng Liang and Haiming Ye. The first draft of the manuscript was written by Rufeng Liang and all authors commented and edited on previous versions of the manuscript. The research was supervised by Weiwen Zhang.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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ENDNOTES

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[ii] https://www.cfi.cn.

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